

Electric field computation in `sourceField1d`

Anna Kelbert with Claude

September 2026

Abstract

`sourceField1d` (EARTH/FWD/field1d.f90) computes both the magnetic field $\mathbf{H}$ (poloidal) and the electric field $\mathbf{E}$ (toroidal) of the 1D layered-sphere solution, from a single scalar toroidal potential $T_l(r)$. $\mathbf{H}$ and $\mathbf{E}$ are built from opposite vector spherical harmonic (VSH) patterns of the same potential, related by Faraday's law. This note derives that relation, gives the exact formulas and grid staggering `sourceField1d` implements, and records the one physical convention choice (radial direction) that distinguishes this solver from its companion, `field1d.s2.f90`.

1 Poloidal and toroidal decomposition

Any smooth vector field $\mathbf{F}$ on a spherical surface splits into three orthogonal parts:

$$\mathbf{F} = F_r \hat{\mathbf{r}} + \nabla_a P + \hat{\mathbf{r}} \times \nabla_a T. \quad (1)$$

$\nabla_a P$ (the tangential gradient of a scalar P) is the **poloidal** part; $\hat{\mathbf{r}} \times \nabla_a T$ (the curl of a radial vector) is the **toroidal** part. For a magnetotelluric source, $\mathbf{H}$ is poloidal ($H_r \neq 0$) and $\mathbf{E}$ is toroidal ($E_r = 0$, purely tangential) — the standard TE-mode picture of deep-Earth induction.

In vector spherical harmonics (degree l , order m), with `vsharm` computing $Y(l, m+1) = Y_l^m$, $Y_t(l, m+1) = \partial Y / \partial \theta$, $Y_p(l, m+1) = (\partial Y / \partial \phi) / \sin \theta$:

$$\Psi_{lm} = \nabla_a Y_{lm} \rightarrow (Y_t \hat{\boldsymbol{\theta}} + Y_p \hat{\boldsymbol{\phi}}) \quad \text{POLOIDAL pattern} \quad (2)$$

$$\Phi_{lm} = \hat{\mathbf{r}} \times \nabla_a Y_{lm} \rightarrow (-Y_p \hat{\boldsymbol{\theta}} + Y_t \hat{\boldsymbol{\phi}}) \quad \text{TOROIDAL pattern} \quad (3)$$

$\mathbf{H}$ uses Ψ_{lm} (theta component Y_t , phi component $+Y_p$); $\mathbf{E}$ uses Φ_{lm} (theta component $-Y_p$, phi component $+Y_t$) — the same pair (Y_t, Y_p) , swapped and one sign flipped. This is the fundamental Maxwell duality for the TE mode:

$$\nabla \times \mathbf{E} = i\omega\mu_0 \mathbf{H} \text{ (Faraday)}, \quad \nabla \times \mathbf{H} = \sigma \mathbf{E} \text{ (Ampère, quasi-static)}. \quad (4)$$

Faraday's toroidal curl gives a poloidal field, and vice versa.

2 The potentials, and the H field

`sourcePotential` returns, at every radius and degree:

`Tnsp = d(Tnr)/dr` — the physical radial derivative of `Tnr`.

Array	Value	Radii	Drives
Tnr (k ,1)	$T(r)$	$R_r(1:N_r)$, cell centers	H_r , both E components
Tnsp (k ,1)	dT/dr	$R_s(1:N_r+1)$, cell faces	H_θ , H_ϕ

sourceField1d assembles, per component (Fortran coefficient ordering $m = 0, +1, -1, +2, -2, \dots$):

$$H_x(i, j, k) [\phi] = \sum_l Y_p(l, \cdot) \text{coef1 Tnsp}(k, l) R_0^2/R_s(k)/l(l+1) \quad (5)$$

$$H_y(i, j, k) [\theta] = \sum_l Y_t(l, \cdot) \text{coef1 Tnsp}(k, l) R_0^2/R_s(k)/l(l+1) \quad (6)$$

$$H_z(i, j, k) [r] = \sum_l Y(l, \cdot) \text{coef1 Tnr}(k, l) R_0^2/R_r(k)^2 \quad (7)$$

(H_r carries no $l(l+1)$ denominator — consistent with how **Tnr**/**Tnsp** are normalized in **sourcePotential**.) Each l -sum is built directly from the $m = 0$ term plus each $+m/-m$ pair (§4) — no post-hoc conjugation of any kind.

3 Electric field: derivation and formulas

Start from $\nabla \times \mathbf{E} = i\omega\mu_0\mathbf{H}$, $E_r = 0$. The theta-component of Faraday's law gives

$$-\frac{1}{r} \frac{d(rE_\phi)}{dr} = i\omega\mu_0 H_\theta = i\omega\mu_0 \text{Tnsp}(r) \frac{R_0^2}{r l(l+1)} Y_t. \quad (8)$$

Integrating (**Tnsp** = $d\text{Tnr}/dr$):

$$E_\phi = -i\omega\mu_0 \text{Tnr}(r) \frac{R_0^2}{r l(l+1)} Y_t. \quad (9)$$

The phi-component gives, by the same steps,

$$E_\theta = +i\omega\mu_0 \text{Tnr}(r) \frac{R_0^2}{r l(l+1)} Y_p. \quad (10)$$

$E_r = 0$ throughout (TE mode). The radial component of Faraday's law is automatically satisfied — not an independent constraint — since E_θ/E_ϕ were built from the SAME potential $T_l(r)$ whose value (**Tnr**) and derivative (**Tnsp**) already solve one consistent radial ODE; the poloidal/toroidal formalism guarantees the two are compatible by construction.

Result:

$$E_\theta = +i\omega\mu_0 \text{Tnr}(r) R_0^2/(r l(l+1)) Y_p \quad (11)$$

$$E_\phi = -i\omega\mu_0 \text{Tnr}(r) R_0^2/(r l(l+1)) Y_t \quad (12)$$

$$E_r = 0 \quad (13)$$

E uses the existing **Tnr** array directly — no separate potential is needed.

Component	H pattern	H potential	E pattern	E potential
theta (H_y/E_y)	$+Y_t$	Tnsp at R_s	$+Y_p$	$i\omega\mu_0$ Tnr at R_r
phi (H_x/E_x)	$+Y_p$	Tnsp at R_s	$-Y_t$	$i\omega\mu_0$ Tnr at R_r
radial (H_z/E_z)	$+Y$	Tnr at R_r	0	—

Angular patterns compared

E_{tang} pairs with H_r (same potential $T(r)$, different angular pattern); H_{tang} pairs with dT/dr (**Tnsp**). This is the Faraday-Ampère duality.

4 Coefficient assembly: the $\pm m$ pairing

Fortran coefficient ordering differs from MATLAB's:

Physical m	Fortran coef1 index	MATLAB amp index
$m = 0$	1	1
$m = +m$	$2m$	$m+1$
$m = -m$	$2m+1$	$l+1+m$

Each degree- l sum adds the $m = 0$ term directly, then each $+m/-m$ pair via the Condon-Shortley identity $Y_l^{-m} = (-1)^m \overline{Y_l^m}$:

$$C = \frac{Y(l, m+1) \text{coef1}(2m) + (-1)^m \overline{Y(l, m+1)} \text{coef1}(2m+1)}{l(l+1)} \quad (14)$$

for the $l(l+1)$ -normalized components (H_θ , H_ϕ , E_θ , E_ϕ); H_r uses the same pairing without the $l(l+1)$ divisor. E_x 's accumulated pair carries an overall leading minus sign, matching the $-Y_t$ pattern above. The $(-1)^m$ factor is required: omitting it reproduces the physical sum only for even m .

No component is conjugated after assembly — each l -sum, once the $\pm m$ pairing above is applied term-by-term, is already the correct physical field.

5 Radial (source-amplitude) normalization

The formulas in §2–3 are written directly in terms of **coef1** — the coefficient array as consumed natively by **sourceField1d** — and **Tnr/Tnsp**, the radial potential/derivative solved once per degree l , independent of m and of the coefficient values. Because **Tnr/Tnsp** never depend on **coeff**, every formula above (H_r , H_θ , H_ϕ , E_θ , E_ϕ) is exactly **linear in coef1** — a coefficient's absolute scale enters every field component identically.

field1d.f90 natively expects **coeff**(l, m) = 1 to mean a **unit external surface radial field**: $H_r(R_0) = Y_l^m(\theta, \phi)$ — the **RADIAL_SURFACE** convention in **EARTH/FWD/output_convention.f90**. Three other conventions are supported for source coefficients authored differently: an amplitude of the external potential term $T_l^{\text{ext}}(r) = \alpha_l^T (r/R_0)^{l+1}$ (**RADIAL_MULTIPOLE**, **field1d.s2.f90**'s own native convention, unit $\alpha_l^T = 1$); a literal r^{l+1} coefficient (**RADIAL_DIMENSIONAL**, Sun & Egbert's own text convention, r in metres); or the classical external geomagnetic potential's Gauss coefficient (**RADIAL_POTENTIAL**, $V = R_0 \sum_l (r/R_0)^l \varepsilon_l^m Y_l^m$, used for ionospheric-current/Sq sources).

To use coefficients authored in one of these other conventions with the formulas in §2–3, rescale them **per degree** l (the same factor for every m of that degree, before substituting as `coefl`):

$$\text{coefl_for_formula}(l, m) = \text{coeff_in_other_convention}(l, m) \cdot \frac{A(\text{SURFACE}, l)}{A(\text{other_convention}, l)} \quad (15)$$

with the per-degree amplitude $A(l)$, relative to the MULTIPOLE baseline (identical to `output_convention.f90`'s `radial_amplitude()`):

radial_norm	$A(l)$
MULTIPOLE	1
SURFACE	$R_0^2/(l(l+1))$
DIMENSIONAL	R_0^{l+1} (overflows double precision for $l \gtrsim 44$)
POTENTIAL	$R_0^2/(l+1)$

Because this rescale is a single real scalar per degree l , applied identically wherever `coefl` appears in §2–3, it multiplies H_r , H_θ , H_ϕ , E_θ , E_ϕ all by the **same** factor for that degree — an overall amplitude change only. It never touches the angular pattern ($Y/Y_t/Y_p$), the relative H/E component ratios, or the impedance $Z = E/H$ (amplitude-independent by construction). This linearity is exactly why rescaling the source coefficients before the solve (`rescale_source_radial`) and rescaling the finished H/E output by the same factor afterward are equivalent operations.

6 Grid staggering

H_r/E 's **Tnr** live at cell-center radii R_r ; H_θ/H_ϕ 's **Tnsp** at cell-face radii R_s (k loops over $N_{rr} = N_r$ vs. $N_{rs} = N_r + 1$ respectively).

Component	theta eval	phi eval	radii	j range
H_x (phi)	node $\theta(j)$	mid $\phi + d\phi/2$	R_s	$2..N_t$
H_y (theta)	mid $\theta + d\theta/2$	node $\phi(i)$	R_s	$1..N_t$
H_z (r)	node $\theta(j)$	node $\phi(i)$	R_r	$j_1..j_2$ (pole-aware)
E_y (theta)	node $\theta(j)$	mid $\phi + d\phi/2$	R_r	$j_1..j_2$ (pole-aware)
E_x (phi)	mid $\theta + d\theta/2$	node $\phi(i)$	R_r	$1..N_t$
E_z (r)	—	—	—	zero

Each E component shares its angular stagger with the H component using the **same VSH factor** — E_y (Y_p) with H_x (also Y_p , node theta/mid phi), E_x (Y_t) with H_y (also Y_t , mid theta/node phi) — not with the H component it pairs with physically via Faraday's law. E_y , not E_x , is therefore the pole-adjacent (node-theta) component and carries the pole-aware j_1, j_2 range: $j_1 = 1$ unless $\theta(1)$ is a true pole ($j_1 = 2$ then); $j_2 = N_t + 1$ unless $\theta(N_t + 1)$ is a true pole ($j_2 = N_t$ then). This range is only ever narrowed for a genuinely global grid reaching both poles — a regional grid keeps its full node range. `node_P_lm` (shared with H_z) is reused for E_y ; `edge_P_lm` (shared with H_y) is reused for E_x — no separate Legendre computation is needed for E .

7 Scope and the radial-direction convention

$E_r = 0$ is exact for the TE (poloidal- H) source this solver targets; a source with TM-mode content would need separate treatment (not needed for standard Sq/ring-current MT sources).

`field1d.f90`'s raw output above is natively $e^{-i\omega t}$, with $\hat{\mathbf{r}}$ pointing away from Earth's center (increasing radius) — identical, in every bookkeeping respect but one, to the companion solver `field1d_s2.f90`. This was a deliberate choice, not the solver's original one: `field1d.f90` was written in 2011 as a faithful implementation of Kelbert (2006)'s own conventions (the Global3D Fortran code in ModEM), which use $\hat{\mathbf{r}}$ pointing toward Earth's center and $e^{+i\omega t}$ output — genuinely different, equally valid physical conventions. `field1d.f90` was later re-pointed to match `field1d_s2.f90` so the two independently-derived solvers agree natively with no compensating sign in comparison code; Kelbert (2006)'s original convention remains available as the named `KELBERT2006` preset in `EARTH/FWD/output_convention.f90`. See `docs/output_conventions.pdf` for the full convention system and the derivation of exactly which field components that radial-direction choice affects.